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REVIEW

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Exploring the mathematic equations behind the materials science data using interpretable symbolic regression

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Abstract

Symbolic regression (SR), exploring mathematical expressions from a given data set to construct an interpretable model, emerges as a powerful computational technique with the potential to transform the "black box" machining learning methods into physical and chemistry interpretable expressions in material science research. In this review, the current advancements in SR are investigated, focusing on the underlying theories, fundamental flowcharts, various techniques, implemented codes, and application fields. More predominantly, the challenging issues and future opportunities in SR that should be overcome to unlock the full potential of SR in material design and research, including graphics processing unit acceleration and transfer learning algorithms, the trade-off between expression accuracy and complexity, physical or chemistry interpretable SR with generative large language models, and multimodal SR methods, are discussed.

KEYWORDS

explainable machine learning, material database, materials science, representation learning, symbolic regression

1 | INTRODUCTION

Materials science, an interdisciplinary field that combines principles from physics, chemistry, and engineering, stands at the forefront of technological innovation, supporting advancements in fields ranging from aerospace, automotive, and electronics to biomedicine research.[1–8] Traditionally, material development has been a time-consuming and resource-intensive process, often characterized by trial-and-error experimentation.[9–14] However, the advent of computational modeling methods, ranging from density functional theory (DFT) and molecular dynamics to finite element simulations, has revolutionized this landscape, enabling the prediction and optimization of material properties and behaviors without the need for extensive physical trials. Despite their successes, traditional computational methods often strive with limitations such as computational cost, the complexity of material systems, and the need for extensive domain knowledge to interpret results effectively.[15–17]

With the remarkable evolution of computer hardware and advancements in artificial intelligence, the paradigm of scientific discovery has shifted from theoretical simulations to a data-driven approach,[18–21] bringing the development of groundbreaking technologies in computer vision,[22,23] deep learning,[24–26] and generative models,[27–30] among others. In materials science, inspired by the initiatives of the Materials Genome Initiative,[31–33] a variety of high-throughput calculations and multi-scale simulation methods have emerged as crucial tools for material design, consisting of Artificial Learning and Knowledge Enhanced Materials Informatics Engineering (ALKEMIE),[34] Pymatgen,[35–37] Atomate,[38] Automated Interactive Infrastructure and Database (AiiDA),[39] Automatic Flow (AFLOW-π),[40,41] and Atom simulation environment (ASE)[42] et al. These simulation methods generate "Big Data", which is estimated in the order of zettabytes and grows by approximately 40% annually.[43–49] To address the challenges of data management, characterized by the 5Vs[50] (volume, velocity, variety, variability, and value), multi-type and multi-modal databases have been constructed, including Inorganic Experimental Databases,[51] Materials Project,[35,36] AFLOW,[52,53] Materials Cloud,[54] NOMAD,[55] COD,[56] and ALKEMIE Cloud[57] et al. Based on the vast amounts of data sets, machine learning (ML), a branch of artificial intelligence, has become a predominant role in materials science[58,59] due to its ability to learn materials behaviors and trends from the "ocean of data,"[60] without relying on underlying physical principles, including many key tools, like TensorFlow,[61] PyTorch,[62] scikit-learn, Matminer,[63] and ALKEMIE-PotentialMind,[64,65] et al.

For material design, the most crucial step is to explore the relationship between the input material descriptors and the goal properties and to construct accurate and extensible models for properties prediction or inverse material design.[66,67] These models, typically based on statistical and probabilistic principles in mathematics, are often considered "black box" models due to their lack of direct physical or chemical interpretability. Symbolic regression (SR),[68–70] which explores mathematical expressions from a given data set, emerges as a powerful computational technique with the potential to enhance material science research. Unlike traditional ML methods, which fit data to pre-specified models, SR autonomously searches for both the form and parameters of the best-fitting models, which allows for the discovery of underlying patterns and relationships in data that might be missed by conventional ML methods, such as Bayesian Network, Support Vector Machine, Random Forests, and so on.

This review focuses on the theories, implementations, applications, and future prospects of SR. In Section 2, we introduce the representation learning and generative models related to SR, along with the theories and methodologies of SR itself. The third section outlines various SR methods based on different ML approaches, benchmark data sets, and implemented codes. Section 4 discusses the application of SR in the selection of material features or descriptors, the prediction of material properties, and atomic interaction potentials. Finally, it looks forward to the current challenges and future opportunities of SR as proposed by the authors.

2 | FROM REPRESENTATION LEARNING TO SR

From 2000 to 2022, there has been a notable increase in the number of scientific papers published on the topic of SR, as indexed in the Science Citation Index and shown in Figure 1. SR is a critical area in ML that focuses on discovering interpretable mathematical models from data. Its development has been marked by several milestones, which are highlighted by contributions from various researchers. In the early 1990s, genetic programming (GP) for SR laid the foundation for this field, which was one of the first methods used to automatically create mathematical models that fit given data sets. Next, the ML-based SR methods, like the Bayesian or random forest methods, began to be applied more widely across various fields, from physics to finance, demonstrating its versatility and power in model discovery without pre-specified model structures. In the 2000s, with deep learning architectures marked a significant advancement, the deep learning SR methods enabled end-to-end training through backpropagation, allowing for the extraction of interpretable models from complex data sets. In recent years, the creation of a generative transformer model and graph model specifically designed for SR represents the latest advancement; the development of physics-inspired algorithms for SR, such as artificial intelligence (AI) Feynman, also demonstrated the potential of combining domain knowledge with ML to uncover complex physics equations from data. This growth reflects an extending academic interest and expanding research activities in the field of SR, aligning with the advancements in computational techniques and data-driven methodologies.

2.1 | Representation learning

ML has seen exponential growth in interest and application over the past few decades. The evolution of ML is marked by the transition from knowledge-driven approaches, where rules were manually encoded into systems, to data-driven approaches, which rely on statistical methods to learn patterns from data. The advent of big data and advancements in computational power have significantly

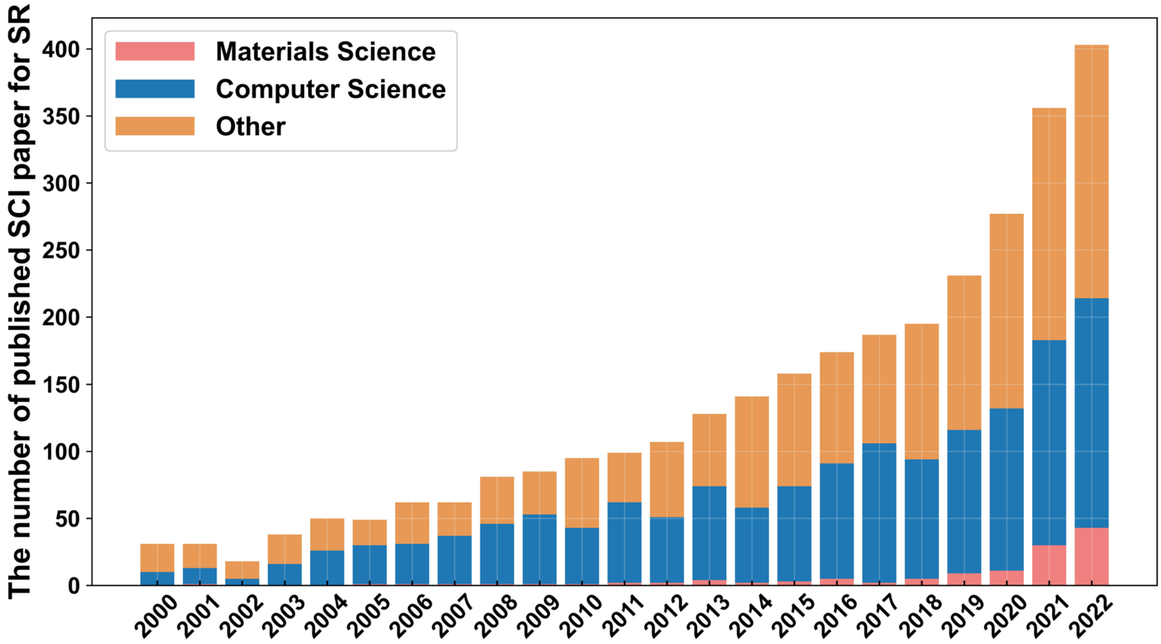

FIGURE 1 The number of published scientific papers on the topic of SR. The original data are queried by the keyword of SR from Web of Science from 2000 to 2022, as indexed in the SCI. SCI, Science Citation Index; SR, symbolic regression.

fueled the growth of ML, allowing for the analysis of large and complex data sets. It can be broadly categorized into several types based on the learning approach, including supervised learning, unsupervised learning, semi-supervised learning, reinforcement learning, representation learning, transfer learning, and ensemble learning.[24] As is well known, the success of ML algorithms often depends on the quality of the input data representation. Driven by feature engineering, hand-crafted features, although effective in some cases, require domain-specific knowledge and can be time-consuming and labor-intensive.[66] In contrast, representation learning[71,72] is a process that aims to automatically discover meaningful representations from raw or unstructured data in an unsupervised or semi-supervised, even generative manner, making it less reliant on human expertise and prior knowledge. The goal is to transform the initial data into a more informative and robust structured format using deep learning models like autoencoders,[73,74] Boltzmann machines,[75] neural networks, or SR. By bridging the gap between raw data and meaningful representations, representation learning can be used to extend new applications and improve the generalizability of ML models.

2.2 | Generative models

Generative models represent a sophisticated category within ML, characterized by their ability to generate new data instances that resemble the training data.[27] In contrast to discriminative models, which focus on categorizing or predicting labels given certain inputs, generative models learn the underlying distribution of the data, enabling them to produce novel data points. The foundational theory behind generative models involves understanding and modeling the probability distribution of data. Generative models, such as Generative Pre-Trained Models (GPTs),[76] Generative Adversarial Networks (GANs),[77–81] Variational Autoencoders (VAEs),[82,83] and Bayesian networks,[84] have been used in diverse applications like image and text generation, style transfer, and protein discovery. Traditional SR constructs mathematical expressions utilizing GP, which involves crossover and mutation processes with refined parameters. In the domain of generative models, this generation of symbolic expressions can be substituted with advanced generative models, which are capable of producing a diverse array of mathematical models and iteratively refining them for optimal representation of input data sets. The details of the implemented software will be presented in Section 3.1.4.

2.3 | Theories and fundamental flowcharts of SR

The mathematical process of SR can be understood through its reliance on GP. This begins with an initial, randomly generated population of candidate models, often represented as binary tree structures. As shown in

Figure 2A, the nodes of these trees symbolize mathematical operations (like +, −, ×, ÷, sin, cos, log, etc.), and the leaves represent variables (x, y, z) and constants (a, b, c).

In the evolutionary algorithm of SR, the model evolves through a series of steps, as shown in Figure 2B:

- ➢ Selection: Models are evaluated for their fitness, typically using a metric such as mean squared error. The fitness function often includes a complexity penalty to prevent overfitting, favoring simpler models.
- ➢ Crossover: This genetic operator combines parts of two-parent models to create new models by exchanging their sub-trees, as shown in Figure 2C.
- ➢ Mutation: This introduces variation by altering parts of the model, such as changing an operation or a variable in the tree structure, as shown in Figure 2D.

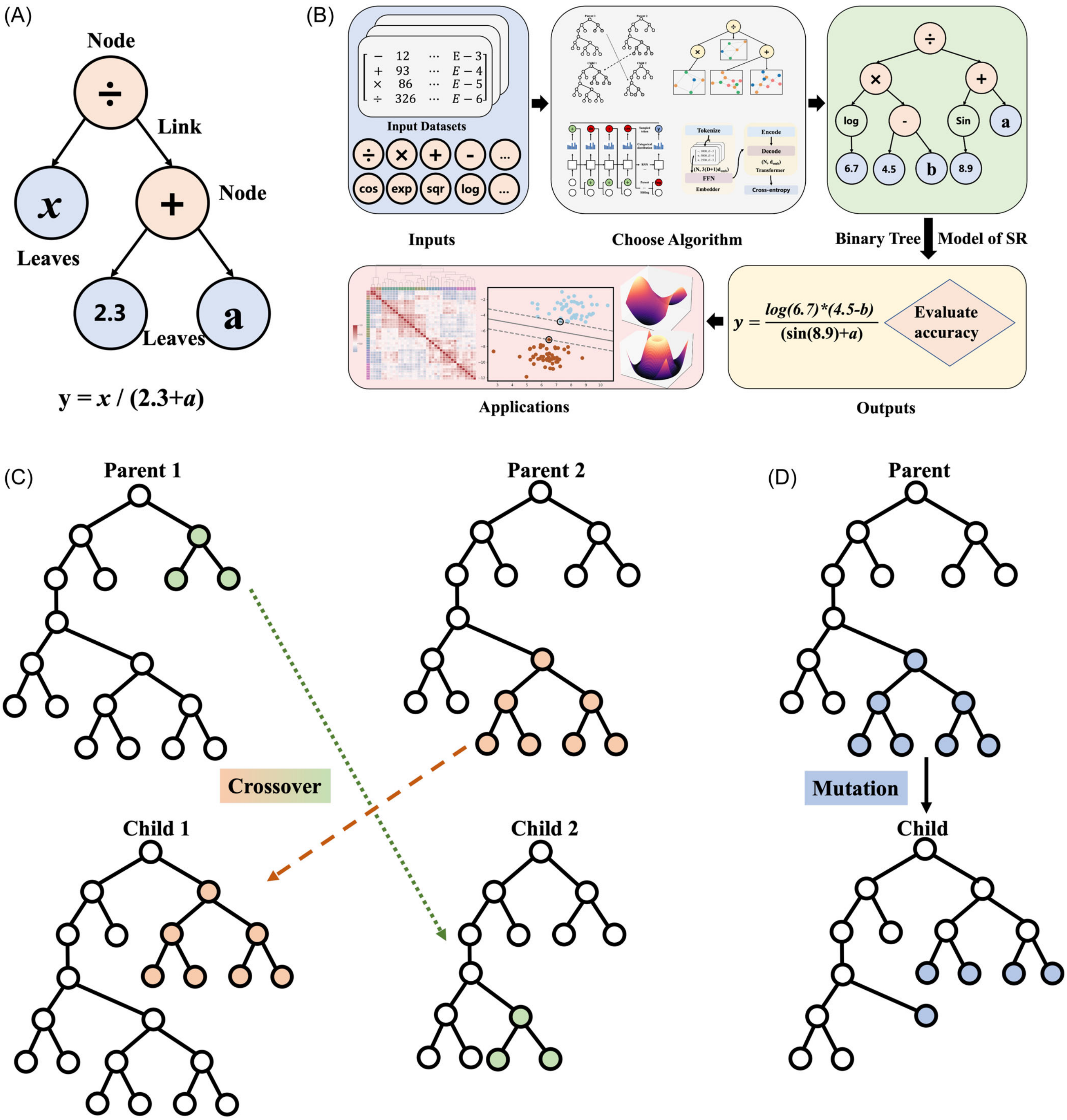

FIGURE 2 (A) The binary tree of SR. (B) The flowchart of general SR. (C) Crossover, and (D) mutation genetic operations in SR. SR, symbolic regression.

➢ Reproduction: The fittest models are selected to generate offspring, forming the next generation of models.

This iterative process aims to minimize an objective function, often the error between the model predictions and actual data, balanced against model complexity. Over generations, the algorithm ideally converges to a model that accurately represents the underlying patterns behind the data.

3 | VARIOUS APPROACHES, BENCHMARKS, AND SOFTWARE OF SR

3.1 | Various approaches of SR

SR methods can automatically analyze the importance of input features and generate appropriate mathematical expressions. As shown in Table 1, these methods are categorized based on the approaches of exploring the underlying expressions, including genetic programming (GPSR), traditional machine learning (TMLSR), deep learning (DLSR), generative-based transformer (TSR), and graph SR (GraphSR).

3.1.1 | Genetic programming SR

Derived from the evolutionary computing method known as genetic algorithms (GA), GP[153–155] is generally used to evolve the best values for a given set of model parameters and allows the basic structure of the model to evolve, along with the values of its parameters, as shown in Figure 3A.[88] The common techniques of GP include Linear GP, Stack-based GP, Cartesian GP, and Positional GP, both with classic tree-shaped mathematic representation.[156]

The first SR method, using object-oriented GP (in C++), was reported by Pyhnen,[85,86] which sought to find the best set of parameters for a pre-specified equation and confirmed the feasibility of this method, but the precision and efficiency of this method are not mentioned. Subsequently, Douglas[87] et al. proposed a more efficient read's linear code to find the best equation coefficient after 3060 iterations, which only consumed 50 min. Furthermore, a new method named Geometric Semantic GP (GSGP)[88] is superior in performance but often results in overly complex, humanly unintelligible expressions. To constrain the complexity of final expressions, another interesting method, named Prioritized Grammar Enumeration (PGE),[95] diverges from typical GP techniques by expanding the best results to more complex expressions, without the requirement of random number generation operators. Additionally, SymTree[100,101] represents a different approach, using a tree structure with a root node containing a linear regression of its "children." There are also many efficient GPSR methods, like generalized SR,[102] TIPSR,[91,97] model-based GP,[92,94] operon C++,[93] Cartesian GP,[90] coefficient mutation SR,[96] and so on, which utilize a greedy heuristic for child node generation, focusing on linear combinations of expressions to obtain a more interpretable model.

3.1.2 | Traditional machine learning SR

GP-based SR techniques are the most popular methods, offering diverse methodologies for mathematical expression derivation from data sets. However, the computational efficiency and expression complexity need to be further enhanced.[99,104,105,107,117,157] Therefore, there are many hybrid SR methods to solve the above challenges, such as gplearn,[98] nonlinear least squares,[110] policy gradients SR,[122,127] Taylor SR,[114] RILS-ROLS,[113,120] nearest neighbor indexing SR,[103] deterministic/GP SR,[89,158] multi-SISSO,[108] multivariate temporal SR,[159] and dimension synchronous computation (SR-DSC).[119] In particular, Rivero et al. introduce a distinctive SR technique, DoME,[118] which presents a novel, efficient, and mathematical approach with the capability of generating user-friendly and versatile mathematical expressions. A key feature of DoME is the ability to limit the complexity of the generated expressions, making them more compact, interpretable, and analyzable. Furthermore, the expressions generated by DoME are standard equations that can be easily utilized in various programming languages without the need for importing machine learning libraries. They are also adaptable for use in different calculation systems, including spreadsheets, offering a versatile advantage over approaches like Neural Networks. Besides, another novel method using a Bayesian framework[109,112,116] is presented, which leverages a Bayesian framework to effectively incorporate prior knowledge, thus addressing the limitations of traditional GP in terms of interpretability and computational efficiency, as shown in Figure 3B. Additionally, the Symbolic Physics Learner (SPL) method[115] employs an efficient Markov chain Monte Carlo algorithm to sample symbolic trees from the posterior distribution, offering a more memory-efficient solution than GP by eliminating the need to maintain a genome pool.

3.1.3 | Deep learning SR

Deep Learning SR (DLSR), combining deep learning and neural networks with SR, has emerged as a significant area of study. The method of Deep SR (DSR)[127] employs a

TABLE 1 List of software and implementation tools for SR.

Name	Type	Year	Author & Ref
GP with C++	GPSR	1996	Pyhne[85]
Rainfall-Runoff GP	GPSR	1999	Savic[86]
GPSR	GPSR	2000	Augusto[87]
Geometric Semantic GP	GPSR	2012	Moraglio[88]
Deterministic GP	GPSR	2013	Icke[89]
Positional Cartesian GP	GPSR	2018	Wilson[90]
Interaction–Transformation GP	GPSR	2019	Franca[91]
Model-based GP	GPSR	2019	Virgolin[92]
Operon C++	GPSR	2020	Burlacu[93]
Improving Model-based GP	GPSR	2021	Virgolin[94]
Probabilistic Grammars GP	GPSR	2021	Brence[95]
Coefficient Mutation GP	GPSR	2022	Virgolin[96]
Transformation-Interaction-Rational (TIR)	GPSR	2022	Franca[97]
Evolving Tree	Hybrid GPTree	2018	Cava[98]
Greedy Tree	Hybrid GPTree	2018	Olivetti[99]
PS-Tree	Hybrid GPTree	2022	Hengzhe[100]
SymTrees	Hybrid GPTree	2023	Baume[101]
GeneralizedSR	Hybrid GPTree	2023	Tohme[102]
Nearest Neighbor Indexing SR	TMLSR	2010	McRee[103]
Eureqa	TMLSR	2011	Dubčáková[104]
Globally SR	TMLSR	2017	Austel[105]
Exhaustive Search	TMLSR	2018	Igarashi[106]
SISSO	TMLSR	2018	Ouyang[107]
Multi-task SISSO	TMLSR	2019	Ouyang[108]
Bayesian SR	TMLSR	2019	Jin[109]
Nonlinear Least Squares	TMLSR	2019	Kommenda[110]
Policy Gradients SR	TMLSR	2019	Peter[111]
Bayesian Neural Network	TMLSR	2021	Cranmer[112]
Informed Equation Learning	TMLSR	2021	Werner[113]
Taylor SR	TMLSR	2022	HeBaihe[114]
Symbolic Physics Learner	TMLSR	2022	SunFangzheng[115]
Automatic Bayesian SR	TMLSR	2022	Vázquez[116]
Shape-Constrained SR	TMLSR	2022	Kronberger[117]
DoME	TMLSR	2022	Rivero[118]
Dimension Synchronous Computation	TMLSR	2022	Wang[119]
RILS-ROLS	TMLSR	2023	Kartelj[120]
Continued Fraction Regression	TMLSR	2023	Moscato[121]
Deep SR	DLSR	2019	Petersen[122]

TABLE 1 (Continued)

Name	Type	Year	Author & Ref
Scaled Fast NN	DLSR	2020	Costa[123]
AI Feynman	DLSR	2020	Udrescu[124]
AI Feynman 2.0	DLSR	2020	Udrescu[125]
Neura SR	DLSR	2021	Biggio[126]
RNN-DSR	DLSR	2021	Petersen[127]
Hierarchical Reinforcement Learning	DLSR	2021	XuDuo[128]
DeepNN SR	DLSR	2021	Kim[129]
Neural-Guided GPSR	DLSR	2021	Mundhenk[130]
Recurrent Sequences DSR	DLSR	2022	Ascoli[131]
Singlet Fission SR	DLSR	2022	LiuXingyu[132]
Controllable NN SR	DLSR	2023	Bendinelli[133]
Control Variables NN SR	DLSR	2023	ChuXieting[134]
Physically Plausible NN SR	DLSR	2023	Kubalík[135]
Smooth TSR	TSR	2015	Pitzer[136]
Grammar Variational Autoencoder	TSR	2017	Kusner[137]
Distorted Video TSR	TSR	2021	Udrescu[138]
SymbolicGPT	TSR	2021	Valipour[139]
Multi-Objective SR	TSR	2021	Kubalík[140]
Symformer	TSR	2022	Vasti[141]
NP-hard SR	TSR	2022	Virgolin[142]
Automated Adaptive	TSR	2022	Narayanan[143]
End-to-end TSR	TSR	2022	Kamienny[144]
Symbolic Expression Transformer	TSR	2022	LiJiachen[145]
TSR for Differential Equations	TSR	2023	Becker[146]
Deep Generative SR	TSR	2023	Holt[147]
Online SR	TSR	2023	Jin[148]
Monte Carlo Tree DGSR	TSR	2023	Kamienny[147]
Joint Supervised Learning TSR	TSR	2023	LiWenqiang[149]
Variational Auto-encoder SR	TSR	2023	Popov[150]
Transformer-based Planning SR	TSR	2023	Shojaee[151]
Deep GNNSR	GraphSR	2020	Cranmer[152]

neural network to search for symbolic expressions. As shown in Figure 3C, they use risk-aware reinforcement learning where a recurrent neural network outputs a distribution over mathematical expression. In addition, another algorithm of AI Feynman's[124,125,128] two-step approach starts with problem simplification, employing techniques like dimensional analysis and translational symmetry, followed by a polynomial fit and brute-force search for complex expressions using a six-layer feed-forward neural network. Furthermore, Igarashi et al. also developed the Exhaustive Search Method[106] and defined a semantically unique expression search space to perform an exhaustive search for the best-fitting expression. Moreover, continued fraction regression (CFR)[121] presented a method using a memetic algorithm with continued fractions, focusing the search on the coefficient space of the CFR. Besides, integrated with controllable neural networks,[123,126,129–134] Kim et al. have shown the Equation

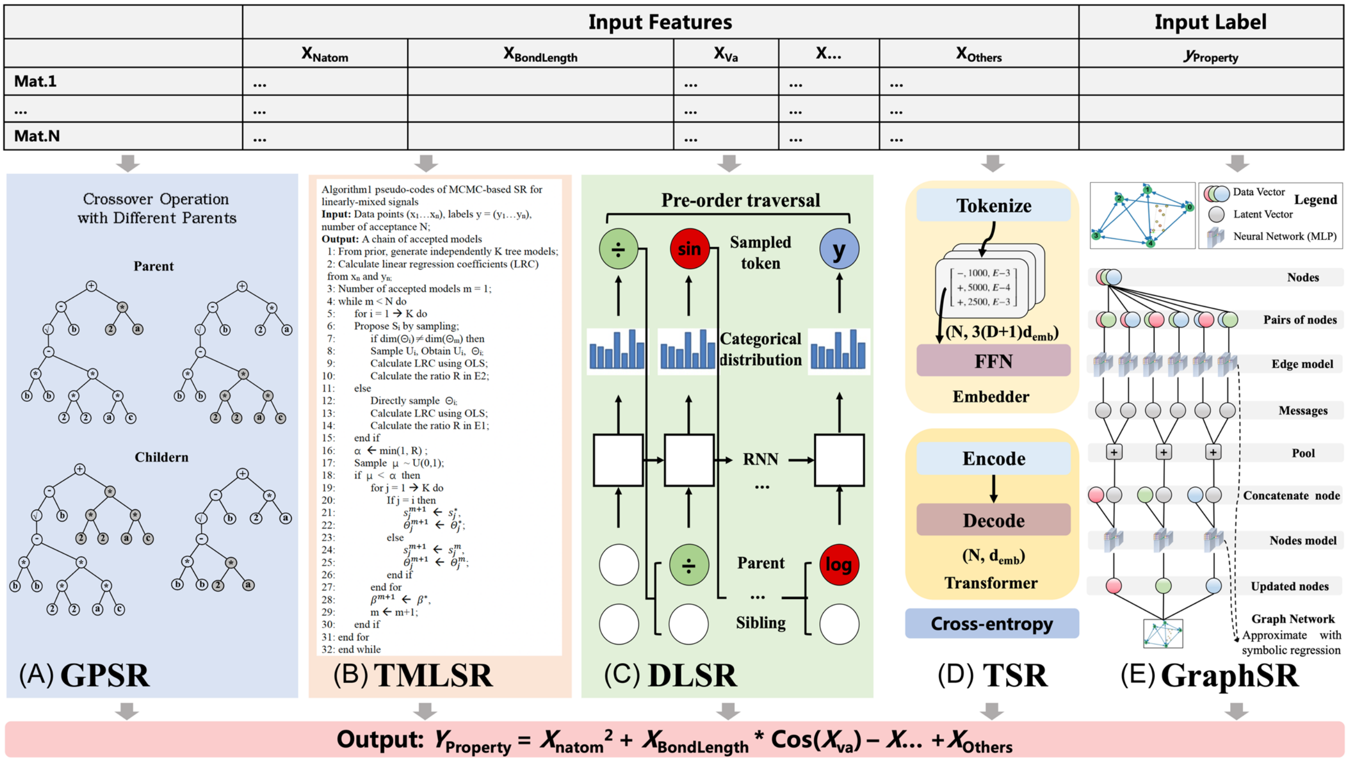

FIGURE 3 The illustration of (A) genetic programming SR (GPSR). (B) Traditional machine learning (TMLSR). Reproduced under terms of the CC-BY license.[109] Copyright 2019, the authors, published by ArXiv. (C) Deep Learning SR (DLSR). Reproduced under terms of the CC-BY license.[127] Copyright 2021, the authors, published by ArXiv. (D) Transformer SR (TSR). Reproduced under terms of the CC-BY license.[144] Copyright 2022, the authors, published by ArXiv. (E) Graph SR. Reproduced under terms of the CC-BY license.[152] Copyright 2020, the authors, published by ArXiv.

Learner network,[135] which enhances the extrapolation and prediction of unfamiliar data. In summary, DLSR represents a powerful integration of deep learning and SR, with methodologies ranging from reinforcement learning to exhaustive search with neural networks. These approaches enhance the ability to discover and interpret complex mathematical expressions, paving the way for advances in data analysis and scientific discovery.

3.1.4 | Generative SR

With the aim of discovering interpretable mathematical expressions from data, Generative SR, an advancing area within machine learning, combines GPSR with generative modeling techniques.[146–148,160] Kusner et al. first reported a variant autoencoder, Grammar-VAE,[137] used for generating expressions based on predefined grammar. The characteristics of this model are the integration of neural networks and symbolic operators as activation functions. In 2022, with the explosive popularity of ChatGPT, the Transformer[161] technology behind it became a hot research topic in various fields. Subsequently, SymbolicGPT,[139] a transformative language model that takes advantage of the strengths of GPT-like models, is reported, demonstrating superior accuracy, speed, and data efficiency. The approach innovatively combines language models with symbolic mathematics, offering a scalable, cost-effective solution with fewer input restrictions and enhanced performance in various SR tasks. Consequently, as shown in Figure 3D, Meta AI presents a novel approach using a Transformer model that directly predicts full mathematical expressions,[144] including variables and constants. This method outperforms traditional GP and other deep learning models in speed and accuracy, showing promise for real-time inference applications and setting a foundation for future transformer-based symbolic tasks. Meanwhile, another method named SymFormer is reported by Martin et al.,[141] which employs a transformer-based approach to efficiently predict both the symbolic representation and the values of all constants in a single neural network pass. This novel method surpasses current state-of-the-art approaches in speed, demonstrating rapid inference capabilities ideal for real-time applications. Furthermore, SymFormer's initial constant predictions for gradient descent optimization enhance its overall performance and effectiveness in uncovering complex mathematical relationships from extensive formula collections. In addition, various automated adaptive SR methods, like smooth SR,[136] automatic video discover SR,[138] multi-objective SR,[140] deep learning SR,[142] automated adaptive SR,[143] joint ML SR,[149] auto-encoder SR,[150] and transformer-based planning SR,[151] are developed. However, the model's adaptability is currently limited by the fixed number of dimensions and sampling range set during training, a constraint that future iterations could address by varying the sampling distribution of input points to increase flexibility across different data sets.

3.1.5 | Graph SR

Graph SR (GraphSR)[152] represents a pioneering approach in machine learning, particularly in distilling symbolic representations from deep models with a focus on Graph Neural Networks (GNN), as shown in Figure 3E. This method starts by training a GNN to encourage sparse latent representations in a supervised setting. SR is then applied to components of the learned model, extracting explicit physical relations such as force laws and Hamiltonians. Further advancements in GraphSR involve imposing physically motivated inductive biases on both general and Hamiltonian Graph Networks, enhancing their ability to learn interpretable representations and potentially improving zero-shot generalization. Remarkably, this technique has been successful in extracting correct known equations from neural networks and even discovering new analytic formulas, as demonstrated in a complex cosmology example involving dark matter simulation. The symbolic expressions derived from the GNN show superior generalization to out-of-distribution data compared to the traditional GNN methods, showcasing the potential of GraphSR in uncovering novel physical principles.

3.2 | Benchmark data sets of SR

Benchmark data sets for SR are crucial in evaluating the performance of various algorithms and methods. These data sets encompass a range of problems, from real-world data sets to ground-truth benchmark problems.[162–165] One notable data set is named AI Feynman,[124] which consists of 100 equations from the Feynman Lectures on Physics. Another data set includes 131 known chaotic dynamical systems,[166] spanning fields such as astrophysics, climatology, and biochemistry, providing a challenging and interpretable benchmark for forecasting models and SR algorithms. Additionally, various methods have been evaluated on these data sets, including Hybrid approaches combining GAs with reinforcement learning and other techniques, which have demonstrated competitive performance compared to standard SR methods.[167] Furthermore, recent advancements like the Symbolic Expression Transformer (SET)[145] use an image caption model to translate images to symbolic expressions, as shown in Figure 4A. Additionally, a large SR benchmark repository called SRBench includes metadata describing the underlying equations, units, and various summary statistics. The average test performance and inference time of different SR models with baselines provided by the SRbench benchmark, both on black-box regression problems and Feynman SR problems, are shown in Figure 4B,C. The average test performance and formula complexity of different SR models with baselines provided by the SRbench benchmark, both on black-box regression problems and Feynman SR problems, are shown in Figure 4D,E.

4 | APPLICATION OF SR IN MATERIALS SCIENCE

4.1 | Selection of material features or descriptors

The selection of material features or descriptors is crucial in materials science as it fundamentally determines the accuracy and effectiveness of predictive models in understanding and optimizing material properties and behaviors. Using gplearn methods of SR, a new descriptor (μ/t) was identified,[168] leading to the discovery and synthesis of five new oxide perovskites, four of which exhibit among the highest intrinsic oxygen evolution reaction activities recorded for this catalyst type. Another study integrates an interpretable SR with ML to formulate new hybrid descriptors that are relevant for the target stability and photovoltage of perovskite films in aqueous conditions.[169] Their approach, combining the random forest algorithm and experimental validation, notably improves the performance of $CH_3NH_3PbI_3$ films with molecular interlayers. Additionally, they develop a t-SR method for designing molecular descriptors, demonstrating its potential in advancing the design

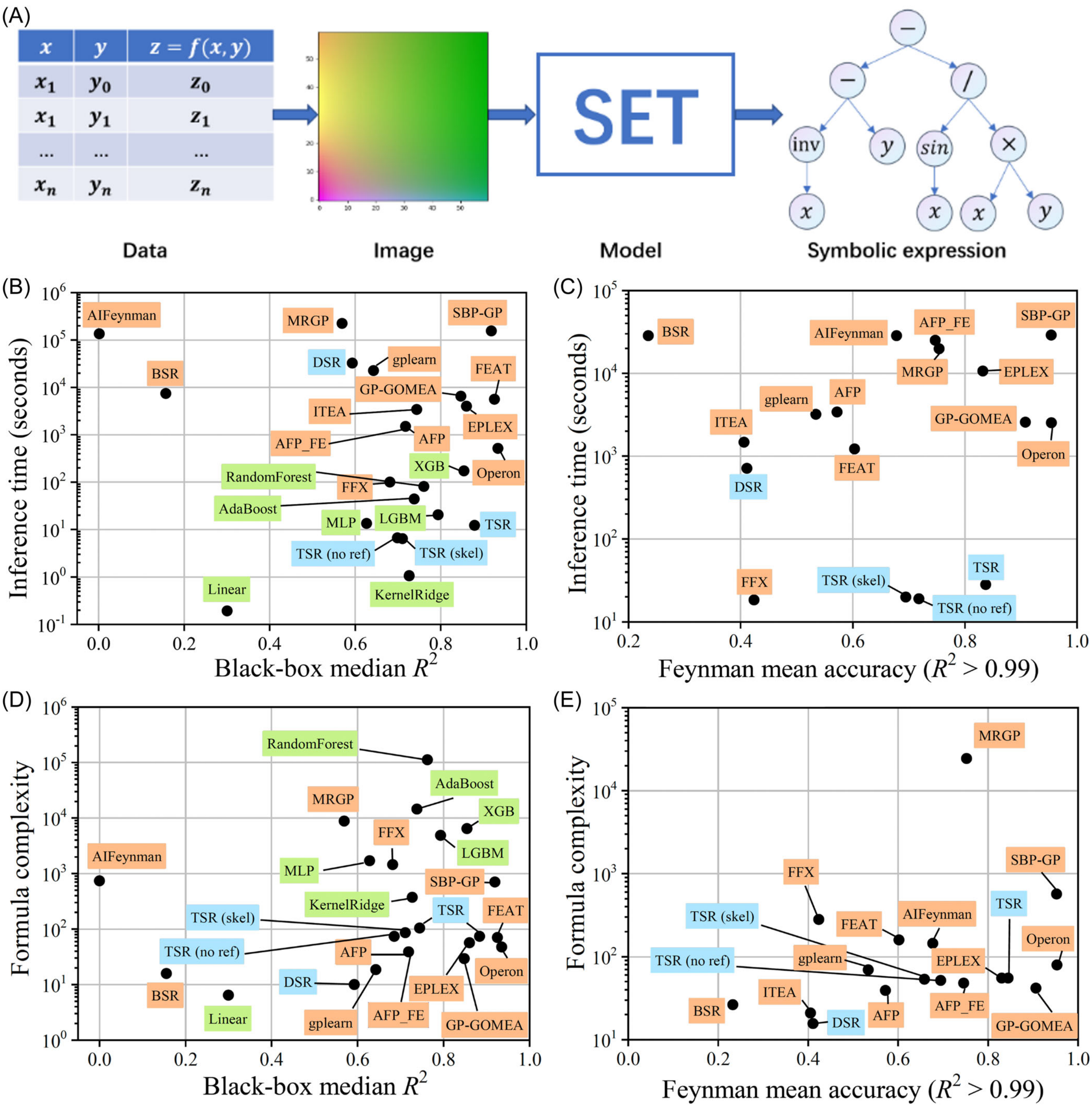

FIGURE 4 (A) The overview of SET. The collected data and the corresponding mathematical expressions are represented as input images and output symbolic sequences, respectively, so that SR is modeled as an image caption task in SET. Reproduced under terms of the CC-BY license.[145] Copyright 2022, the authors, published by ArXiv. (B, C) Pareto plot comparing the average test performance and inference time of different models with baselines provided by the SRbench benchmark, both on black-box regression problems and Feynman SR problems. (D, E) Complexity-accuracy Pareto plot. Pareto plot comparing the average test performance and formula complexity of different models with baselines provided by the SRbench benchmark, both on black-box regression problems and Feynman SR problems. (B–E) are reproduced under the terms of the CC-BY license.[144] Copyright 2020, the authors, published by ArXiv. SET, Symbolic Expression Transformer; SR, symbolic regression.

of robust, high-performing perovskite films for industrial applications. Moreover, Wenguang et al.[170] also highlight the efficacy of the random forest SR algorithm in achieving high accuracy and identifying key descriptors like polarizability and bond type for adsorption energy.

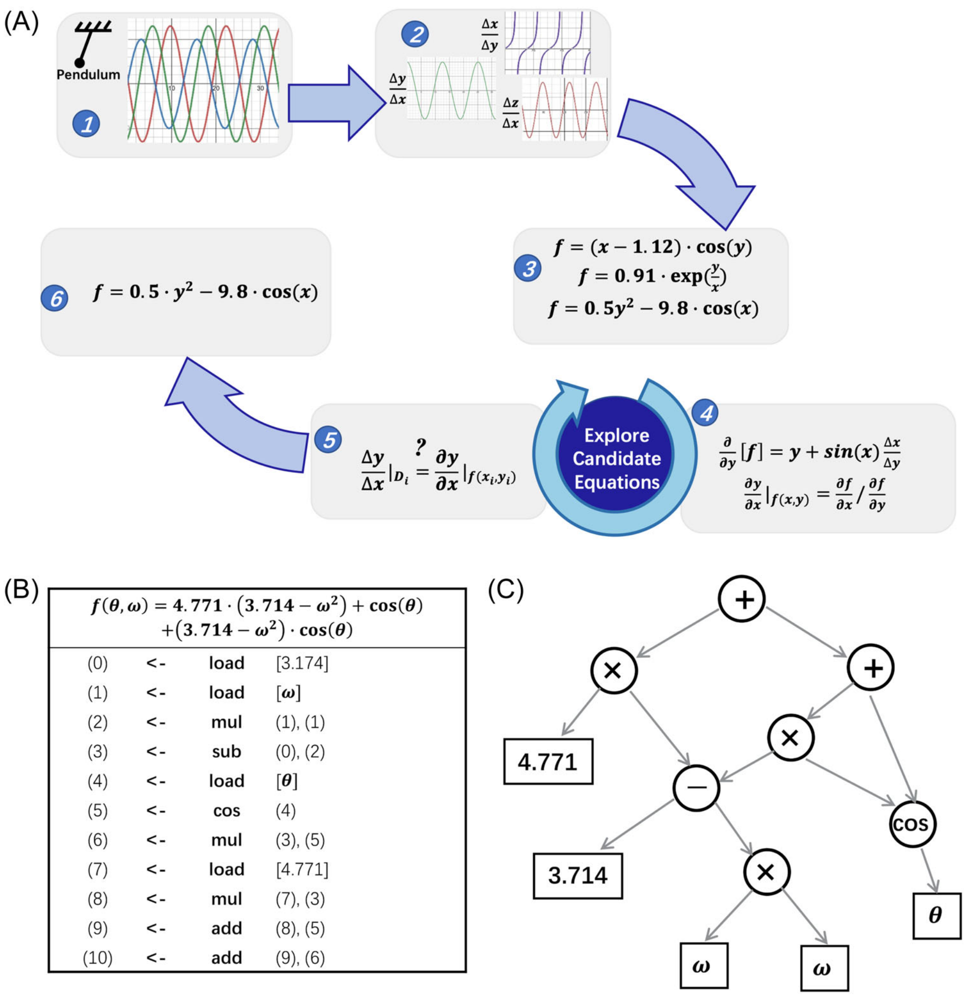

FIGURE 5 Computational approach for detecting conservation laws from experimentally collected data. (A) First, calculate partial derivatives between variables from the data, then search for equations that may describe a physical invariance. To measure how well an equation describes an invariance, derive the same partial derivatives symbolically to compare with the data. (B) The representation of a symbolic equation in computer memory is a list of successive mathematical operations. (C) This list representation corresponds to a graph, where nodes represent mathematical building blocks, and leaves represent parameters and system variables. Reproduced with permission.[175] Copyright 2009, Science.

4.2 | Prediction of material properties

4.2.1 | Structure stability

SR is a powerful tool in the prediction of structural stability. This study[171] demonstrates that the interfacial binding strength between single metal atoms and oxide supports in single-atom catalysts (SACs), essential for preventing sintering, can be predicted using a statistical learning approach based on least absolute shrinkage and selection operator regression, enabling the empirical screening and design of robust catalysts against sintering. Additionally, SR also has been effectively applied to the stability analysis of single-atom alloy catalysts (SAAC). By integrating SR with an iterative variable selection (VS-SISSO) method, the study[168] successfully reduced model complexity while maintaining accuracy. This approach was particularly applied to analyze the segregation energy (SE), a key indicator of SAAC stability, leading to the development of more compact yet accurate models compared to previous high-dimensional ones. Moreover, SR with neural networks is used to discover a new criterion for predicting glass-forming ability.[172] By combining with the "white-box" sure independence screening and sparsifying operator (SISSO) approach, an interpretable and accurate formation energy model[173] used for the prediction and classification of formation energies of binary compounds is constructed. These advancements demonstrate SR's potential in simplifying complex models in materials science without decreasing predictive performance.

4.2.2 | Mechanical properties

For mechanical properties of metal, GPSR is utilized to develop material models with fewer assumptions and high interpretability. Generally, although SR is typically used to find explicit and differential equations for mechanical properties, this method cannot readily find physical laws or equations. Rather than trying to find a certain equation format, this study[174] is trying to detect any underlying physical law that the system obeys, as shown in Figure 5A. First, they collected experimental data from the physical system, and numerically calculated partial derivatives for every pair of variables. Then, they generated random candidate equations by GPSR, like Equation 1.

$$\begin{aligned} f &= (x-1.2)\cdot\cos(x) \\ f &= 0.91\cdot\exp\left(\frac{y}{x}\right) \\ f &= 0.5y^2 - 9.8\cdot\cos(x). \end{aligned} \tag{1}$$

The representations of SR in programming and graphs are shown in Figure 5B,C. To measure how well an equation describes an invariance, they derive the symbolic partial derivatives of pairs of variables for each candidate function $\left(\frac{\partial y}{\partial x} = \frac{\partial f}{\partial x}/\frac{\partial f}{\partial y}\right)$ to compare with the data $\left(\frac{\Delta y}{\Delta x}\right)$. Next, they compared predicted partial derivatives with numerical partial derivatives and selected the best equations. Finally, when predictive ability reaches sufficient accuracy, the most parsimonious Equation 2 with physical laws can be generated.

$$f = 0.5y^2 - 9.8\cdot\cos(x). \tag{2}$$

For porous ductile metals, this study[174,175] focuses on the Gurson model, methodically relaxing its assumptions to evaluate their impact on GPSR model forms. By incorporating transfer learning (TL) methods and regularizing the GPSR fitness function, the research demonstrates that GPSR can generate physically valid, interpretable models, which allows the generation of material models addressing limitations in the Gurson model and conjectures material strength reduction due to voids. The resulting models, more interpretable and applicable than traditional methods, highlight the potential of GPSR combined with TL techniques in materials science. Furthermore, for strain rates ranging from 10^{-4} to 10^{12} (quasi-static to highly dynamic) and temperatures ranging from room temperature to over 1000 K, the flow stress models[176] derived from SR are assessed by comparison to Taylor anvil impact data, showing candidate free-form models comparing well with data in regions of stress-strain space with sufficient experimental data, pointing to a potential means for both rapid prototyping in future model development. In addition, SR methods also are used for the prediction of four mechanical properties[177] of steels—fatigue strength, tensile strength, fracture strength, and hardness, stress–strain curves,[178–180] yield strength of amorphous alloys,[181] the plasticity of single crystal Ni-based superalloys,[182] the heat treatment process of dual-phase steel,[183] the relation between flow stress and temperature-compensated strain rate for an aluminum alloy AA7055,[184] creep formulation,[185] hardness of transition metal compounds,[186] and bulk properties of perovskites.[187]

4.2.3 | Physical properties

The band gap is a crucial topic in materials physics, especially for the performance of perovskite solar cells, as it directly influences their light absorption characteristics. Traditional methods of band gap engineering, such as experimental trial-and-error or high-throughput DFT calculations, tend to be inefficient and costly. In a recent study,[188] SR was combined with feature engineering and the Gradient-Boosted Regression Tree (GBRT) algorithm to develop an interpretable ML strategy for predicting the band gap accurately. Consequently, based on the band gap's affecting factors of the electronegativity difference between the B-site and the X-site (χ_{B-X}), a mathematical formula ($E_g = {\chi_{B-X}}^2 + 0.881\chi_{B-X}$) was constructed with GASR for a quantitative interpretation of the band gap influence patterns. Additionally, another research[189] employed feature augmentation with SR to rapidly estimate and screen inorganic double perovskites with an accurate correlation coefficient of 92.4%. Furthermore, an interpretable Δ-machine learning (Δ-ML) model with SR[190] is constructed to predict the hybrid functional HSE bandgap based on the PBE functional bandgap, which delivers a practical and understandable model for identifying the best physical descriptors for machine learning-based band gap predictions in the extensive exploration of new 2D materials.

For other physical properties, the researchers successfully showcased the framework's effectiveness by creating an accurate analytic model for lattice thermal conductivity[191] with only 75 experimental measurements,[192] which further enabled them to screen 732 materials and identify 80 ultra-insulating materials through hierarchical analysis. Moreover, using high-throughput experiments and machine learning, researchers have discovered a generalizable and accurate expression for ionic transport in battery electrolytes,[193] reflecting fundamental physical mechanisms and aiding in the optimization of electrolyte properties. Besides, other researchers have successfully employed SR to identify and synthesize high-efficiency thermoelectric materials,[194] notably $Cu_{0.45}Ag_{0.55}GaTe_2$, demonstrating a significant figure of merit and highlighting the potential of AI in advancing material science.

4.2.4 | Catalyst properties

Energy and catalytic materials play a pivotal role in advancing technologies like electrocatalysis, crucial for reactions such as the oxygen reduction reaction (ORR) and oxygen evolution reaction (OER). Based on GPSR, we have established the models containing 10 intrinsic element features of SACs.[195] The generated symbolic expressions describing the catalytic activity were provided to quickly and accurately identify catalysts with excellent ORR and OER catalytic activity. First, we gathered a total of 117 initial data for SAC systems and divided them into 52.5% for training, 21% for testing, and 25% for validation. Next, 37 features (descriptors) were assigned for each given data point, which can be directly extracted from element characteristics. Through feature engineering, 10 independent features, which include the number of d-orbital electrons (N_e) in transition metal atoms (TM), the covalent bond of the TM element ($rcov_{TM}$), the atomic number of TM (Z_{TM}), and the number of valence electrons in A and B atoms (V_A and V_B), and so on, are chosen. Subsequently, we have integrated the Random Forest method with GP algorithms to construct a SR approach, the detailed architecture and usage step of the SR model are shown on GitHub (https://github.com/ALKEMIE-Lab/sraes). The final equations for predicting ORR and OER properties are shown in Equations 3 and 4. The mean square errors for the aforementioned equations were 0.095 and 0.094, respectively, corresponding to the prediction accuracy of 79.91% and 75.36% for ORR and OER, respectively. Our work provides several novel and promising ORR and OER electrocatalysts and develops efficient models to reasonably guide and design new catalysts. Additionally, other studies[196] selected the electrochemical step symmetry index method to assess OER intermediates to optimize activity and identify the best SACs.

$$\eta^{\mathrm{OER}}_{\mathrm{predicted}} = \sqrt{0.825 + \sqrt{Z_{\mathrm{TM}}} - I_{\mathrm{TM}} - N_e} + \sqrt{\frac{X_{\mathrm{TM}}}{N_B} - N_e}\,, \tag{3}$$

$$\eta^{\mathrm{ORR}}_{\mathrm{predicted}} = \sqrt{\frac{\sqrt{rcov_{\mathrm{TM}}}}{N_e} - N_e + \frac{N_{\mathrm{TM}}}{N_B}}. \tag{4}$$

4.3 | Atomic interaction potential

The main scope of this section is to apply SR to data obtained from experiments or *ab initio* simulations to find interatomic potentials, which can be used to transfer information from one space-time scale to a higher one. Abdel et al.[197] have demonstrated the efficacy of GPSR in efficiently computing empirical interatomic potentials, such as Lennard-Jones and Born-Mayer potentials, using minimal computational resources. The algorithm efficiently converges to accurate results, as validated through examples of well-defined potentials. Additionally, the authors extend its application to *ab initio* interatomic potentials, exemplified by modeling the potential energy of an Ar_2 molecule using DFT calculations, as shown in Equation 5.

$$V(r) = 1906.22 \cdot \left[\frac{r}{5.71} \cdot (1 + e^{0.171021878791222}) + 0.885215174960045\right]^{-12} - 1141.370159. \tag{5}$$

They adapt GPSR to handle complex data, achieving significant accuracy in a remarkably short computation time. Notably, GPSR successfully identifies specific mathematical forms, such as power −12 for short-range repulsive interactions, demonstrating its potential in accurately characterizing physical interactions from a diverse range of data sources. In addition, Alberto et al.[198] have developed a machine-learning algorithm based on GPSR to create efficient and accurate Lennard-Jones potential, a Sutton-Chen embedded-atom method potential, and many-body potential models for atomistic simulations. This approach, focusing on simplicity and fundamental physical principles, effectively reduces overfitting and enhances model generalizability. It has been validated by rediscovering established potentials and successfully predicting properties of elemental copper beyond the training data. Recently, Burlacu et al.[68] and Alberto et al.[199] have also developed an efficient SR method for discovering physics-derived functional forms of interatomic potentials and validated it for copper. Future development directions include expanding the capabilities of the framework to include three- or many-body interactions, to consider model derivatives to model atomic forces as well.

5 | CHALLENGES AND OPPORTUNITIES AHEAD

5.1 | GPU acceleration

Graphics processing unit (GPU) acceleration significantly enhances the performance of SR algorithms. For instance, grammatical evolution in SR can achieve up to 39 times faster processing when properly utilizing GPU acceleration.[200] Similarly, using an ensemble of extreme learning machines accelerated by GPUs, regression on large data sets can be executed in a reasonable time, enhancing training and model structure selection.[201] To address the challenges posed by increasingly large material data sets and more complex generative models in the future, it is crucial to develop efficient GPU-accelerated algorithms, which are key to expediting the design of new materials, allowing for faster processing and analysis of complex data. Their implementation will significantly enhance the speed and efficiency of material design in an era where data complexity and volume are rapidly escalating.

5.2 | Transfer learning

In terms of transfer learning, it offers a framework for evolving simpler models with better cross-domain generalization performance in GP-based SR, thereby reducing overfitting.[202] Transfer learning in GP also facilitates the extraction of potentially good and unique building blocks from a source problem, improving the solution of SR problems.[203] This method not only speeds up convergence but also helps in obtaining better final solutions across various scenarios, indicating its potential to enhance the efficiency and effectiveness of SR techniques.[204]

5.3 | The trade-off between expression accuracy and complexity

GA-based SR models can provide interpretability but may have limitations in accuracy. As mathematical expressions become more intricate, balancing a model's interpretability and accuracy becomes increasingly difficult. SR inherently grapples with the richness of mathematical expressions,[139,205] necessitating models that strike a balance between interpretability and precision. Multi-objective SR approaches allow for an automatic adaptation of complexity, thereby enabling a balance between accuracy and complexity.[206] Additionally, methods like the Bayesian SR technique improve interpretability and efficiency by incorporating prior knowledge and employing symbolic trees for more concise expressions.[109] This delicate balance between interpretability and accuracy remains a crucial consideration in the advancement of SR methodologies.

5.4 | Physical or chemistry interpretable SR boosted by large language models (LLMs)

SR in the context of LLMs presents both challenges and opportunities for automatic learning of the physical and chemical significance of symbols. SR methods like fast function extraction, GP, and physics-inspired algorithms have demonstrated the ability to learn analytically tractable models from data, effectively predicting the temporal evolution of dynamical systems and discovering complex physics equations from numerical data.

5.5 | Multimodal SR

Material data, with its diverse range encompassing numerical values, 3D configurations, experimental spectra, images, and texts, present a complex array of data types. The task of constructing a SR method that is adept at handling such multimodal data poses both significant challenges and opportunities. This endeavor requires innovative approaches to integrate and interpret these varied data formats effectively. The development of such a method holds the potential to significantly advance the field of materials science, offering more comprehensive insights into material innovations.

6 | CONCLUSIONS

In conclusion, this paper explores the theoretical underpinnings of SR, contrasts it with traditional regression models, and highlights its unique advantages and challenges. We also delve into recent methodological advancements, consisting of computational software and benchmark data sets. Furthermore, a comprehensive overview of how SR can be applied to various aspects of materials research is provided from the selection of significant features to the prediction of material properties and atomic interaction potential. Finally, a critical analysis is carried out to identify current limitations and propose future research directions. This paper underscores the application potential of SR in materials science. By offering a novel computational approach that combines predictive power with interpretability, SR stands to significantly accelerate materials discovery and innovation, paving the way for new technologies and applications.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

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